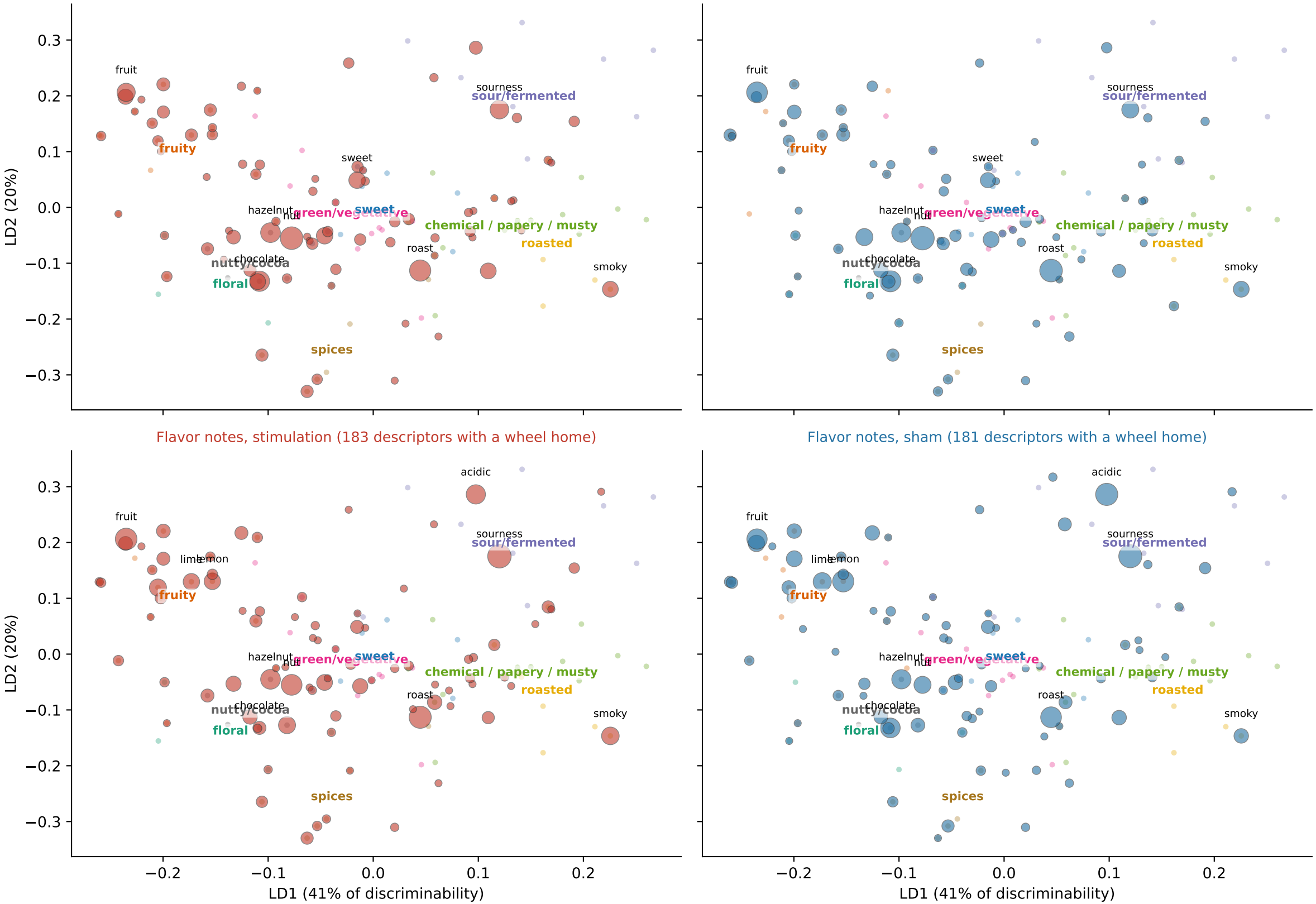
Aroma notes, stimulation (180 descriptors with a wheel home)

Aroma notes, sham (174 descriptors with a wheel home)

Flavor notes, stimulation (183 descriptors with a wheel home)

Flavor notes, sham (181 descriptors with a wheel home)



Flavor-wheel space: wheel terms (small points, coloured by family) and the words participants used that map to the wheel (large points; area = occurrences in the panel's condition, same scale in all panels; red = stimulation, blue = sham), in the plane (LD1, LD2) that best separates the wheel families (discriminant analysis fitted on the wheel vocabulary only; Qwen3-Embedding-0.6B); green/vegetative, sweet and nutty/cocoa overlap in this plane. Descriptive map; the 2D projection distorts distances.
Tests (family endorsement, paired sign-flip permutation over 9 wheel families; exploratory): aroma omnibus p = 0.018, q = 0.035 (cosine-mapping sensitivity q = 0.057); flavor omnibus p = 0.54, q = 0.54; no single family reaches q < 0.25 (BH over 18 family tests). Differences in the map are not tests of individual words.